

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-306 — GEN

Time: Three hours

Maximum Marks: 70

*Instructions: 1, Attempt any two out of four questions in Section A.***SECTION — A**

Long-Answer Questions. Answer any two questions:

2 × 12 = 24 Marks**Question 1.** Explain 3-layer and 5-layer IoT architectures in detail.**Question 2.** Differentiate between microprocessors and microcontrollers. What are the application areas of microprocessors and microcontrollers?**Question 3.** What is the purpose of sensors in IoT? What criteria are required to choose a suitable sensor to measure a desired physical parameter?**Question 4.** What do you understand by the term IoT Ecosystem? What are the components of an IoT system? Describe all.**SECTION — B**

Attempt any six questions:

6 × 6 = 36 Marks**Question 5.** What do you mean by IoT? What are the advantages and disadvantages of IoT applications?**Question 6.** What is the cloud? What are the basic characteristics of cloud computing?**Question 7.** Differentiate between RISC and CISC microprocessors.**Question 8.** What is the difference between transducers, sensors, and actuators?**Question 9.** What do you mean by an Embedded System? Explain its characteristics.**Question 10.** Explain the components of a microcontroller.**Question 11.** Define the following terms:

- (a) Data Processing
- (b) Automation
- (c) Communication Protocols

Question 12. Explain any six applications of IoT.

Question 13. Discuss the principles of operation for each type of sensor and their applications in IoT devices.

SECTION — C

Question 14. Attempt all parts:

10 × 1 = 10 Marks

- (a) Which protocol provides Full-Duplex communication?
 - (a) WebSocket
 - (b) HTTP
 - (c) MQTT
 - (d) CoAP
- (b) The RFID technology consists of:
 - (a) Antenna
 - (b) Transponder
 - (c) Transceiver
 - (d) All of these
- (c) Connected car is:
 - (a) BAN
 - (b) LAN
 - (c) WAN
 - (d) None of these
- (d) The full form of LoRa is:
 - (a) Long Range
 - (b) Long Radio Range
 - (c) Low Range
 - (d) None of these
- (e) Which of the following wireless technologies are used in cellular networks?
 - (a) GSM
 - (b) CDMA
 - (c) LTE
 - (d) All of the above
- (f) Which of the following is a lightweight protocol used in IoT applications?
 - (a) MQTT
 - (b) WebSocket

- (c) HTTP
 - (d) All of the above
- (g) Which of the following wireless technologies provides mesh network architecture in the easiest way?
- (a) Bluetooth
 - (b) ZigBee
 - (c) Wi-Fi
 - (d) None of these
- (h) What is the full form of HTTP?
- (a) Hyphenation Text Test Program
 - (b) Hypertext Transfer Protocol
 - (c) Hypertext Transfer Package
 - (d) None of the above
- (i) An entity in IoT:
- (a) Businesses
 - (b) Governments
 - (c) Consumers
 - (d) All of the above
- (j) ZigBee is based on which of the following standards:
- (a) IEEE 802.15.4
 - (b) IEEE 802.5
 - (c) IEEE 802.4.13
 - (d) Both (a) and (b)